A bounded counterexample to pointwise convergence for recurrence-defined functions

MathDB problem 354674

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Abstract

Pointwise convergence of the scaled coefficients of a backward recurrence does not imply a continuously differentiable limit, even when the limiting differential equation has a unique solution continuous at the terminal endpoint. We give an explicit counterexample with recurrence values in $[0,1]$, and a variant with convex-combination coefficients.

Task 1. Full resolution

Conjecture 1 of Bayón et al. [1], retained in the final publication at pp. 29–30, is MathDB problem 354674. It concerns

$$F_n(k) = G_n(k) + H_n(k)F_n(k+1), \qquad F_n(n) = \mu.$$

Put $f_n(x) = F_n(\lfloor nx \rfloor)$, $g_n(x) = nG_n(\lfloor nx \rfloor)$, and $h_n(x) = n(1 - H_n(\lfloor nx \rfloor))$. The hypotheses are pointwise convergence $g_n \to g$, $h_n \to h$ on $(0,1)$ with continuous limits, and solvability of $y' = -g + hy$ continuously up to 1 only for one terminal value $y(1) = \Theta$. The conjectured conclusion is pointwise convergence of f_n to a function in $C^1(0,1]$ satisfying that equation and terminal condition. The following example disproves it.

Theorem 1. *The conjecture is false even with $\mu = \Theta = 0$, $0 < H_n(k) < 1$ for $k < n$, and $0 \le F_n(k) \le 1$. In fact, h_n can converge locally uniformly on $(0,1)$.*

Proof. For every $n \ge 1$ and $0 \le k \le n$, define

$$H_n(k) = \frac{n-k}{n-k+1}.$$

For $n \ge 4$, set

$$j_n = \lfloor n/2 \rfloor + 1, \qquad G_n(k) = \begin{cases} 1, & k = j_n, \\ 0, & k \ne j_n. \end{cases}$$

For $n < 4$, set $G_n(k) = 0$ for all k. Set $F_n(n) = 0$ and determine $F_n(k)$ backward by the recurrence. The harmless value at $k = 0$ makes all floor-index expressions defined on $[0,1]$; restriction to $\{1, \dots, n\}$ gives the domain stated in the conjecture.

For $n \ge 4$ we have $j_n < n$. Backward substitution gives $F_n(k) = 0$ for $k > j_n$, $F_n(j_n) = 1$, and, for $k < j_n$,

$$F_n(k) = \prod_{\ell=k}^{j_n-1} \frac{n-\ell}{n-\ell+1} = \frac{n-j_n+1}{n-k+1}.$$

Consequently

$$F_n(k) = \begin{cases} \dfrac{\lceil n/2 \rceil}{n-k+1}, & 0 \le k \le j_n, \\ 0, & j_n < k \le n. \end{cases} \tag{1}$$

These values lie in $[0,1]$. For $n < 4$, F_n is identically zero.

Fix $x \in (0,1)$. If $x \le 1/2$, then $\lfloor nx \rfloor \le \lfloor n/2 \rfloor < j_n$. If $x > 1/2$, then $\lfloor nx \rfloor > j_n$ for all sufficiently large n, since $j_n/n \to 1/2$. Thus $G_n(\lfloor nx \rfloor) = 0$ eventually for every fixed x, including $x = 1/2$, and $g_n(x) \to g(x) = 0$. Also

$$h_n(x) = \frac{n}{n - \lfloor nx \rfloor + 1} \longrightarrow h(x) = \frac{1}{1-x}.$$

Writing $n - \lfloor nx \rfloor + 1 = n(1-x) + d_n$ with $1 \le d_n < 2$ gives $0 < h(x) - h_n(x) \le 2/[n(1-x)^2]$, which proves local uniform convergence of h_n. Both limiting coefficients are continuous, indeed smooth, on $(0,1)$.

The limiting equation is $y' = y/(1-x)$. Hence

$$\big((1-x)y(x)\big)' = 0, \qquad y(x) = \frac{C}{1-x}.$$

A solution continuous at 1 must have $C = 0$. Thus the only admissible terminal value is $\Theta = 0$ and the only such solution is $y \equiv 0$, which belongs to $C^1(0,1]$. The endpoint-solvability hypothesis therefore holds in full.

On the other hand, (1) gives the pointwise limit

$$\lim_{n\to\infty} f_n(x) = \begin{cases} \dfrac{1}{2(1-x)}, & 0 < x \le 1/2, \\ 0, & 1/2 < x < 1. \end{cases} \tag{2}$$

Its value at $1/2$ is 1 and its right-hand limit there is 0. It is not continuous and cannot be the claimed C^1 function. For example, $f_n(1/4) \to 2/3$, whereas the only admissible differential-equation solution has value zero there. □

Corollary 2. *The conjecture remains false if one also requires* $0 \le G_n(k) \le 1 - H_n(k)$ *and* $0 < H_n(k) < 1$ *for* $k < n$.

Proof. For $n \ge 4$, change only the exceptional coefficients to $G_n(j_n) = H_n(j_n) = 1/2$, and keep the other coefficients as above. Now $F_n(j_n) = 1/2$, so the entire solution in (1) is divided by two. Each fixed x eventually avoids the exceptional index, so the limiting coefficients and endpoint uniqueness are unchanged. The limit is half of (2) and still has a jump. The stated coefficient inequalities follow directly. Equivalently, the recurrence has the form $F_n(k) = (1 - H_n(k))P_n(k) + H_n(k)F_n(k+1)$ with $P_n(j_n) = 1$ and $P_n(k) = 0$ elsewhere. □

Why the hypotheses miss the spike. For every $n \ge 4$, the main example has $\sum_{k=0}^{n-1} G_n(k) = 1$ although $g_n(x) \to 0$ pointwise. The corresponding mass is concentrated on one interval of length $1/n$. The averaged error condition in Theorem 2 of [1] is stronger: for its candidate $f = 0$, the average in condition (3)(iii) is 1 for every $n \ge 5$. Thus that theorem does not assert convergence for this example.

Relationship to earlier counterexamples. Ravsky's 2024 answers [2, 3] show that related recurrence convergence claims can fail because of exceptional boundary coefficients. Their limiting coefficients are $g = h = 0$, so every constant terminal value is possible; they do not satisfy the unique terminal-value hypothesis here. Our example verifies that additional hypothesis and produces an interior jump.

Four follow-up directions

Task 2. Averaged coefficient control. Formulate sufficient local uniform-integrability or discrete L^1 conditions on the scaled coefficients, together with endpoint control, that recover the proposed conclusion.

Task 3. Limits with concentrated mass. Develop a version of the recurrence limit theorem with measure-valued forcing and explicit jump conditions for finitely many coefficient spikes.

Task 4. Endpoint selection. Determine conditions under which bounded solutions of the limiting linear equation automatically extend continuously to the terminal endpoint, and combine these with compactness of the discrete solutions.

Task 5. Quantitative convergence. For a corrected theorem, bound the error on compact subintervals using discrete accumulated coefficient errors and terminal data.

References

[1] L. Bayón, P. Fortuny Ayuso, J. M. Grau, A. M. Oller-Marcén, and M. M. Ruiz, *A new method for computing asymptotic results in optimal stopping problems*, Bull. Malays. Math. Sci. Soc. **46** (2023), article 46. doi:10.1007/s40840-022-01436-4. Preprint: arXiv:2205.08495v1, Conjecture 1, p. 24; final version, pp. 29–30.

[2] A. Ravsky, answer and comment on MathOverflow question 463221, February 11 and 16, 2024. https://mathoverflow.net/questions/463221.

[3] A. Ravsky, answer and comments on Mathematics Stack Exchange question 4862107, February 16–17, 2024. https://math.stackexchange.com/questions/4862107.